Unified College Club Management System

1. Project Overview

The Unified College Club Management System is a full-stack application designed to manage college club activities efficiently. It provides a centralized platform for handling user registration, event management, attendance tracking, and certificate generation with verification.

Technology Stack

- Backend: Spring Boot (Java)
- Frontend: React
- Database: MySQL

2. Development Approach

The project will be developed using an integrated approach where each team member is responsible for implementing their assigned feature completely, including backend APIs, frontend interface, and database design.

This ensures:

- Faster development
- Clear ownership of features
- Better understanding of full-stack development

3. Team Responsibilities

Pawan Joshi

Backend

- Implement user authentication system
    - POST /register
    - POST /login

Frontend

- Develop registration page
- Develop login page
- Handle form validation and API integration

Database

- Design and manage Users table

Fields:

- id
- name
- email
- password
- role

Anushka

Backend

- Implement event management APIs
    - POST /events (create event)
    - GET /events (fetch events)
    - POST /register-event (event registration)

Frontend

- Develop event listing page
- Implement event registration functionality

Database

- Design and manage:

Events table:

- id
- title
- date
- description

Registrations table:

- id
- user_id
- Event_id

Shaurya Singh

Backend

- Implement attendance management system
    - Mark attendance
    - Fetch user-related data for dashboard

Frontend

- Develop dashboard page
- Create attendance interface

Database

- Design and manage Attendance table

Fields:

- id
- user_id
- event_id
- status

Rishabh Jakhmola

Backend

- Implement certificate system
    - Generate unique certificate ID
    - Store certificate data
    - Certificate verification API
        - GET /verify/{certificateId}

Frontend

- Develop certificate viewer page
- Implement certificate verification page

Database

- Design and manage Certificate table

Fields:

- id
- user_id
- event_id
- certificate_id
- issue_date

4. Development Workflow

- Each team member will work on their assigned feature independently
- Backend APIs should be tested before frontend integration
- All code should follow consistent naming conventions
- Changes should be pushed regularly to the repository
- Integration of all modules will be done gradually

5. Expected Outcome

- Fully functional full-stack application
- User authentication system
- Event creation and registration system
- Attendance tracking system
- Certificate generation and verification system
- Clean and user-friendly interface
- Well-structured and maintainable codebase